Tech Nation Visa - Exceptional Promise
Criteria: Optional Criteria 4 (Academic Contributions through Research)
Applicant: Odeyemi Olusegun Israel

Evidence 4: Academic Publications & Global Research Impact

1. Research Output and Early Traction

Four mathematically dense pre-print papers published independently across open-access archives. The two core papers (Figures 1 and 2) recorded a combined **365 views and 210 downloads** without institutional promotion. The remaining two papers demonstrate the upward trajectory of the research programme.

2. The AMTTP Infrastructure Paper (TechRxiv)

Published: 27 February 2026 | **Metrics:** 230 Views, 153 Downloads (as of 5 May 2026)
DOI: `10.36227/techrxiv.177220113.33816607/v1`
Inaugural paper introducing the AMTTP architectural framework for bridging distributed ledger technology with off-chain ML inferences. 230 views and 153 downloads in roughly ten weeks, without institutional promotion.

Figure 1: TechRxiv platform analytics showing 230 views and 153 downloads for the AMTTP infrastructure paper.

3. Blind-Spot Decomposition Theory — BSDT (Zenodo)

Published: 5 March 2026 | **Metrics:** 135 Views, 57 Downloads (as of 5 May 2026)
DOI: `10.5281/zenodo.18870590`
Proposes a geometric framework decomposing ML fraud-detection failures into four measurable components: Camouflage, Feature Gap, Activity Anomaly, Temporal Novelty. 135 views and 57 downloads confirms sustained interest in novel blind-spot correction approaches.

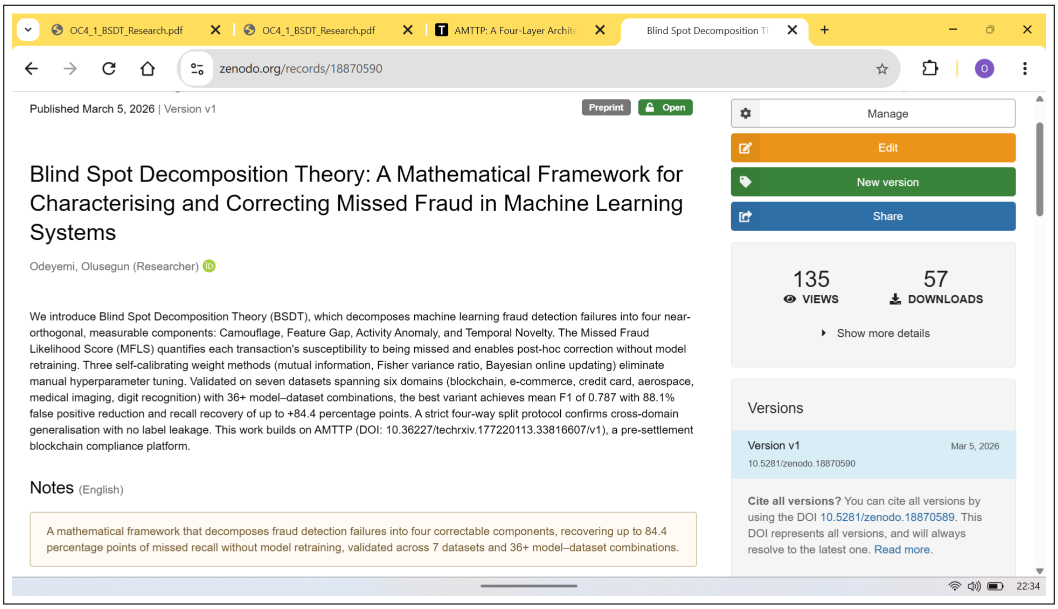

Figure 2: Zenodo platform analytics showing 135 views and 57 downloads for the BSDT mathematical framework.

4. Blind-Spot Decomposition and the Geometry of System Collapse (Zenodo)

Published: 16 March 2026 | **DOI:** `10.5281/zenodo.19038783`

This third paper is the strongest theoretical extension: it reformulates BSDT as a geometric and control problem, shows the energy functional is topologically equivalent to a coupled dynamical system via Morse-index equivalence and homotopy equivalence of sublevel sets, and derives a spectral phase-transition result (adaptive friction required above the critical manifold). Validated cross-domain without retraining: FDIC banking data (AUROC 0.867, six-quarter GFC lead), Terra/Luna collapse (30-hour alarm), and ERCOT Texas grid (AUC 0.9997, 72-hour lead before rolling blackouts). This is evidence of reusable mathematical theory, not just product implementation.

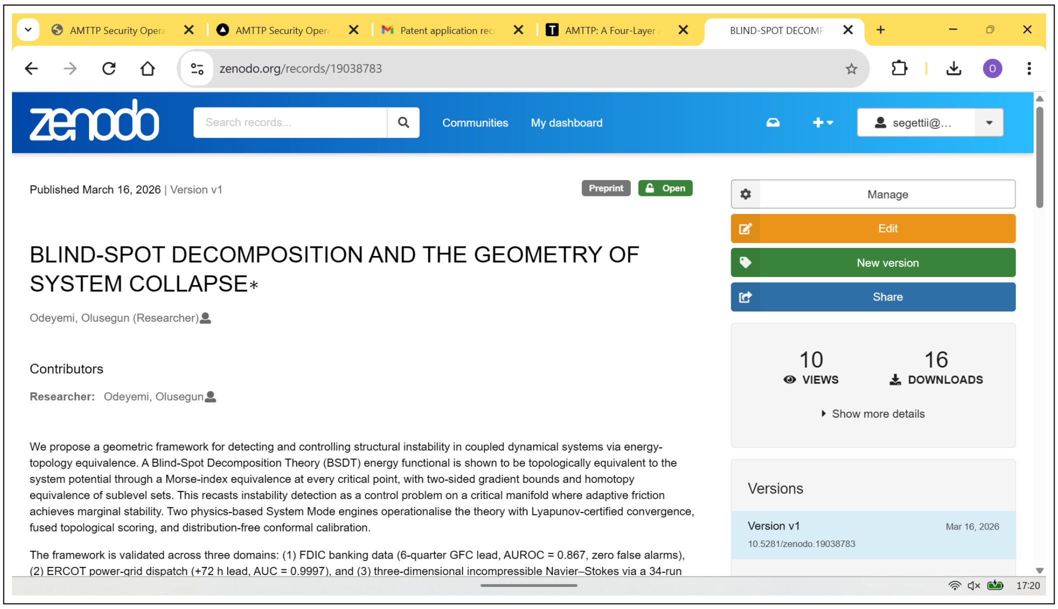

Figure 3: Zenodo analytics for the Geometry of System Collapse paper.

5. Universal Deviation Principle (UDP)

Published: 15 March 2026 | **Version DOI:** `10.5281/zenodo.19037688`
My fourth paper extends BSDT into a broader multi-operator anomaly-detection framework, formalising a tensor-style representation across 14 composable operators from five mathematical families (basis decomposition, geometric embedding, algebraic, topological, information-theoretic) and proving six formal results including global sensitivity via the Extreme Value Theorem and finite-sample guarantees via Hoeffding concentration. I include it to show the upward research arc — from AMTTP architecture, to BSDT failure decomposition, to a general theory of instability and detection.

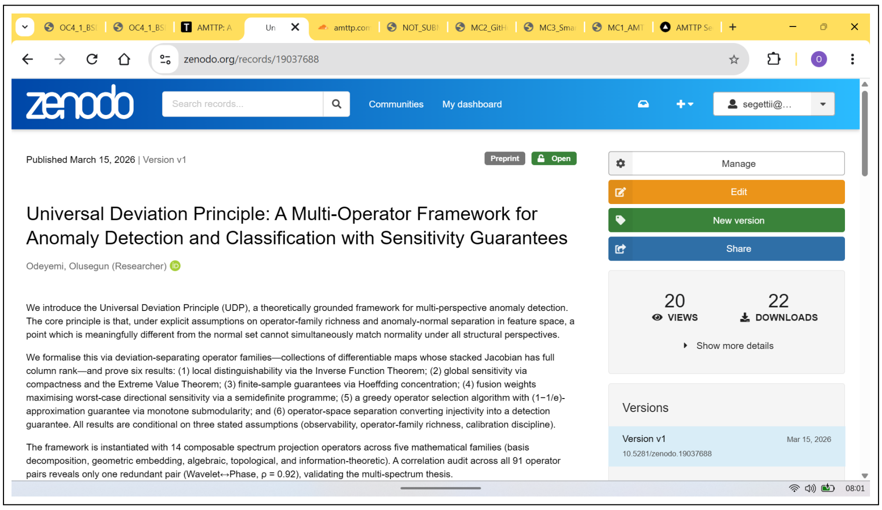

Figure 4: Zenodo record for the UDP paper (20 views, 22 downloads) — a newer output included to evidence continued research output rather than for traction metrics.

6. Independent Academic Endorsement

Two senior academics have reviewed and endorsed this research in signed letters submitted with this application.

Prof. A. S. O. Ogunjuyigbe (Professor of Electrical & Electronic Engineering and Provost, Postgraduate College, University of Ibadan) has mentored the applicant for over ten years and reviewed the work in depth. Writing from his specialism in power systems and computational intelligence, he independently assessed the BSDT/Geometry of System Collapse findings — including the 72-hour early warning (AUC 0.9997) before the ERCOT Texas blackouts — and judged them "*highly significant*" from a domain-expert perspective, noting that the framework derives the same systemic vulnerability identified by the subsequent FERC/NERC investigation, but mathematically in advance, from system data alone.

Dr Sunday Adeola Ajagbe (Senior Lecturer, Abiola Ajimobi Technical University; Research Associate, University of Zululand; *Top 2% World Scientist 2025*) is an independent researcher with no prior supervisory relationship to the applicant. He confirmed in a signed letter dated 2 April 2026 that all three AMTTP-series publications — the UDP, BSDT/Geometry, and AMTTP architecture papers — were adopted as recommended readings in his postgraduate course **CPE 604: Techniques in Machine Learning for Low-Resource Languages**, with a specific reference to the TechRxiv publication page. He describes the work as demonstrating "*originality and seriousness of the thinking*" and "*exceptional promise in digital technology.*" Full supporting course evidence is provided in the separate **OC4 Addendum: CPE 604 Academic Adoption**.